

The Brazilian GDP in Black and White

A portrait of structural inertia through Markov Chains and XGBoost

This analysis employs **Markov Chains** to identify expansion, recession, and stagnation regimes in the Brazilian GDP (1996-2025), validating them against the **Business Cycle Dating Committee (CODACE)**. By integrating the **XGBoost** algorithm with **SHAP Values**, we deepen the investigation of the inertial causality that traps the economy. While Markov's stochastic approach defines the current "state," Machine Learning reveals the "why": a visceral dependence on the immediate past that projects a growth of only **0.43%** for the first quarter of 2026—consolidating an annual ceiling of **1.74%**, significantly below the market's more optimistic expectations.

Analyzing the Brazilian GDP

The data sample comprises the non-seasonally adjusted series from Q1 1996 to Q4 2025. The application of three regimes is justified by the observation of extensive periods in which the Brazilian economy remained in a state of stagnation, as illustrated in **Chart 1**.

Chart 1 – Regime Identification in the Brazilian GDP

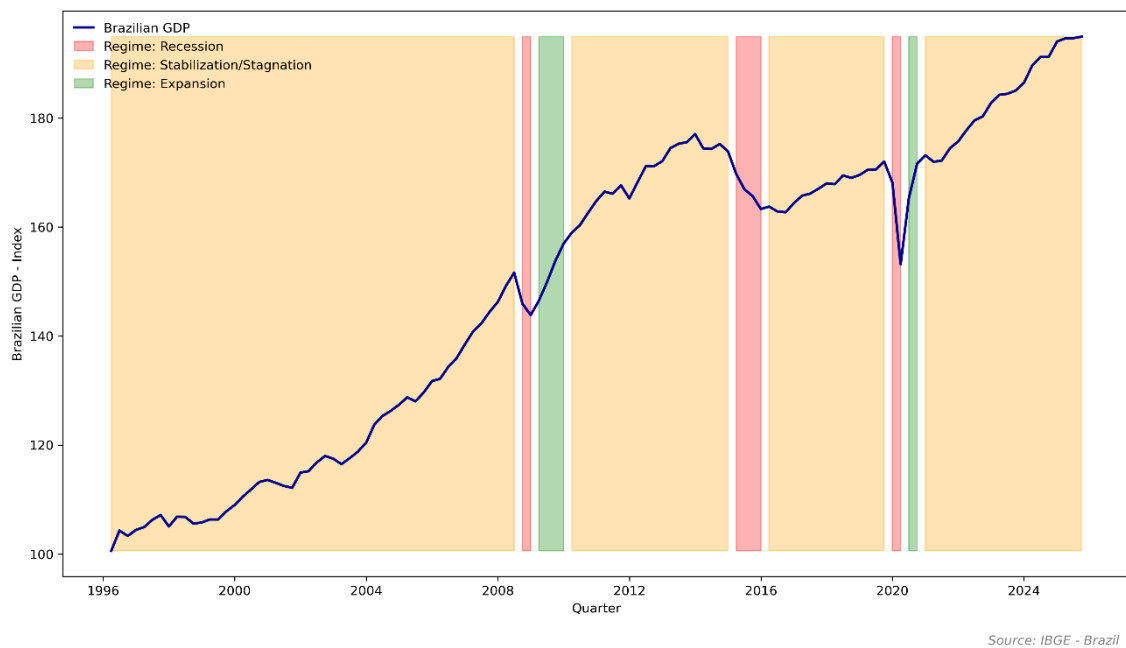


Table 1 details this chronology, showing that stagnation is not a random event but the system's rest pattern.

Table 1 – Chronology of Regimes

Regime	Start	End	Quarters
Stabilization/Stagnation	Q1 1996	Q3 2008	50
Recession	Q4 2008	Q1 2009	2
Expansion	Q2 2009	Q1 2010	4
Stabilization/Stagnation	Q2 2010	Q1 2015	20
Recession	Q2 2015	Q1 2016	4
Stabilization/Stagnation	Q2 2016	Q4 2019	15
Recession	Q1 2020	Q2 2020	2
Expansion	Q3 2020	Q4 2020	2
Stabilization/Stagnation	Q1 2021	Q4 2025	20

Before analyzing the stabilization/stagnation stratum, it is necessary to address the expansionary movements¹. Except for Q2 2015 to Q1 2016—the worst moment of the recent period, fueled by water shocks and a paralysis in government confidence followed by economic inertia—all expansions did not result from organic growth or medium/long-term planning, but rather from crisis mitigation measures:

- **Q2 2009 to Q1 2010: Countercyclical policies** were adopted, such as tax exemptions, Fiscal Policy-driven consumption stimuli, and aggressive credit expansion, primarily through state-owned banks.
- **Q3 2020 to Q4 2020:** Policies such as direct income transfers to families and support programs to maintain liquidity for micro and small enterprises were implemented.

A critical point revealed by the modeling is the nature of the expansion regime. Unlike mature/responsible economies where accelerated growth frequently transitions into sustainable stability, Brazilian data shows that expansion is a transitory state of extremely low persistence. Notably, the probability of the expansion regime transitioning directly into stabilization/stagnation is statistically negligible.

This suggests that Brazilian growth does not "pass through a stability phase"; it is interrupted by shocks that cause it to abruptly converge to the mean or into recession. The **Structural Inertia Map (Chart 2 – Decision Boundaries)** visually illustrates this phenomenon: expansion events are considered dispersed outliers, while the mass of data is sucked into the central low-growth cluster.

Table 2 portrays the coefficients of the different strata of GDP behavior. It demonstrates that the country only reaches expansion levels when the economy undergoes a strong retraction. In other words, the primary goal of expansions is to "put out the fire" of domestic or global contractions. A striking point is that opposing coefficients show very similar magnitudes: while in recession the quarterly average is a decline of **2.31%**, expansion shows an advance of **2.82%**. Added to the residual variance (σ^2) of **0.86**, this statistical noise is not just a leftover; it is the component that sabotages the sustainability of any "chicken flight" in our economy. Stabilization/stagnation ends up being the "tie-breaker," with a mediocre quarterly growth of **0.67%**.

¹ Regarding the recession/contraction case, the analysis can be found in the text [Brazil GDP: Markov Switching Regime Analysis \(1996 – 2025\)](#).

Table 2 – Parameter Coefficients

	Coefficient	P-Value
Constant _{Recession}	-2.31	0.0000
Constant _{Stab/Stag}	0.67	0.0000
Constant _{Expansion}	2.82	0.0000
σ^2	0.86	0.0000

For visualization purposes in the **Structural Inertia Map (Appendix – Chart 2)**, the economy's states of vigor were defined by the following quarterly variation (y) intervals: **Severe Recession**² ($y < -1\%$); **Stagnation** ($-1\% \leq y < 0.5\%$); **Structural Inertia** ($0.5\% \leq y \leq 1.5\%$), and **Real Expansion** ($y > 1.5\%$). Notably, the 0.67% mean identified by the Markov model sits exactly at the heart of Structural Inertia, proving that the sample's critical mass (88.2%) does not merely gravitate around this center but is hermetically contained within a range where growth is insufficient to generate real dynamism.

When dissecting the average rate and volatility by regime and period (**Table 3**), the situation becomes more transparent, highlighting the economy's profound inability to transition between different strata.

Table 3 – Average Rate and Volatility by Regime and Periods

Regime	Start	End	Average Rate (%)	Volatility
Stabilization/Stagnation	Q1 1996	Q3 2008	0.84	1.03
Recession	Q4 2008	Q1 2009	-2.64	1.69
Expansion	Q2 2009	Q1 2010	2.18	0.32
Stabilization/Stagnation	Q2 2010	Q1 2015	0.51	0.96
Recession	Q2 2015	Q1 2016	-1.57	0.67
Stabilization/Stagnation	Q2 2016	Q4 2019	0.35	0.47
Recession	Q1 2020	Q2 2020	-3.56	1.77
Expansion	Q3 2020	Q4 2020	4.29	0.75
Stabilization/Stagnation	Q1 2021	Q4 2025	0.64	0.60

The Record of Stagnation (or Inertia as the *status quo* of the Brazilian Economy)

The Transition Matrix reveals that the probability of the economy remaining in the stabilization/stagnation regime is **96.3%**. In practice, this percentage describes a persistent state of equilibrium where the forces of attraction toward the center of the distribution overcome any vector of dynamism. The mathematical expectation for the duration of this regime suggests that the Brazilian economy does not merely visit stagnation; it settles into it as a rest pattern. Specifically, the stabilization/stagnation period lasts, on average, **27.14 quarters**—approximately 7 years to "jump" to another regime³.

The starkness of this diagnosis is evident when observing the evolution of real growth rates within this state. Between 1996 and 2008, the stability regime showed an average quarterly growth of 0.84%. However, in the most recent economic period (2021-2025), this growth

² The classification of **Severe Recession** ($y < -1\%$) was established to isolate contraction episodes that exhibit statistical relevance compared to the -2.31% constant identified by the Markov model. Values ranging between 0% and -1% are treated by the system as borderline variations, often captured by the **Stagnation zone**, given the high probability of inertial reversal in low-magnitude declines.

³ The recession and expansion regimes last 2.24 and 2.73 quarters, respectively.

retreated to 0.64%, a drop of 0.20 percentage point. This is not just an economy walking slowly; it is a structure showing signs that it is losing the ability to sustain levels previously considered mediocre.

In the Appendix, an exercise was conducted using a Machine Learning tool called XGBoost, which endorses the economy's state of total paralysis. The results showed that the immediately preceding quarter carries a weight nearly 4 times greater than that of two previous quarters, as seen in Appendix Chart 1. Additionally, in Appendix Chart 2, it is recorded in a very forceful manner that the difficulty of breaking away from the inertial level is practically impossible. It is no coincidence that 105 out of 119 data points—88.2% of the entire sample—are found in the region of little or almost no movement, and that the quarterly average of this cluster is 0.67% (see Table 2).

In the same exercise, the forecast for the first quarter of 2026 can be calculated, and it stood at **0.43%**, which was considered a stagnation variation. The figure is obviously overestimated, as it disregards recent global and domestic events: concerning the global sphere, the Middle East conflict, which is significantly affecting oil prices (Brent); and on the domestic front, the clash between the Executive and Legislative branches. This has a potential influence, not only political but economic, as it may affect future investments, reverberating through the National Accounts item, Gross Fixed Capital Formation (GFCF).

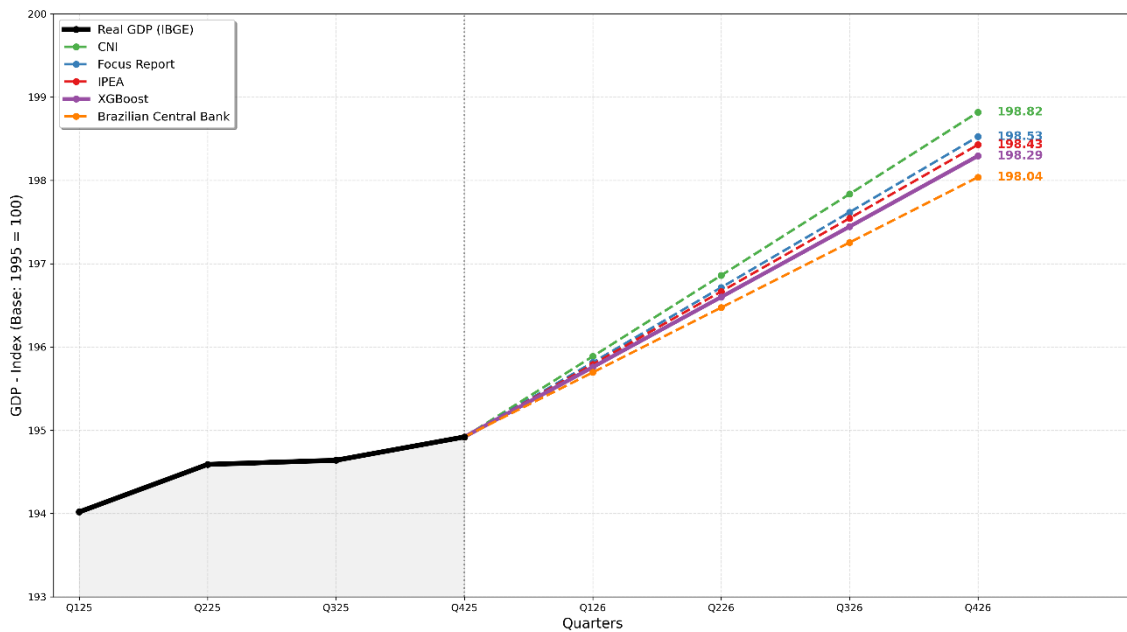
Table 4 – Annual Brazilian GDP Rate Projection (2026)

Institutions	Annual GDP Rate Projection (2026)
National Confederation of Industry (CNI)	2.00%
Focus Report	1.85%
IPEA	1.80%
XGBoost	1.74%
Central Bank of Brazil	1.60%

However, even with this conservative projection, it is not far from the projections of respected institutions when forecasting is mentioned. The table above displays the projections from the leading institutes alongside the one generated by XGBoost⁴.

⁴ For the 2026 GDP prediction, the 0.4314% variation was replicated for each quarter, yielding an annual projection of 1.74%.

Chart 2 – The Clash of Projections: Institutions vs. Structural Inertia



Despite being a "wild projection," the XGBoost forecast was 0.11 percentage point (p.p.) below the market benchmark (**Focus Report**) and, ironically, 0.14 p.p. above the **Central Bank's** forecast.

Resilience, Moderate Optimism, or Mediocrity?

What makes this diagnosis irrefutable is the methodological synergy: two distinct universes—Classical Econometrics (Markov Chains) and Machine Learning (XGBoost)—collided with exactly the same result. While the former models state transitions stochastically, the latter seeks predictive patterns via residual error. Their convergence toward a future of inertial paralysis validates that the Brazilian structure is "addicted" to low altitudes.

The market often labels the current paralysis as "resilience". However, for something to be considered resilient, it must possess the ability to return to its original state after a shock. The data negates this thesis: we observe a structural retreat of 0.20 p.p. in average growth between the historical block (1996-2008) and the current one (2021-2025). Mediocrity is not resilient; it is worsening.

When confronting the projections of the leading institutions for the first quarter of 2026, the convergence toward the center of the distribution is undeniable.

Table 5 – Quarterly GDP Rate Projection

Institutions	Quarterly GDP Rate Projection
National Confederation of Industry (CNI)	0.49%
Focus Report	0.46%
IPEA	0.45%
XGBoost	0.43%
Central Bank of Brazil	0.40%

The variation between CNI's optimism and the Central Bank's conservatism is a mere 0.09 p.p.. The fact that XGBoost, anchored purely in historical inertia, sits in the middle of this dispute suggests that institutional projections are merely pricing in the continuity of mediocrity.

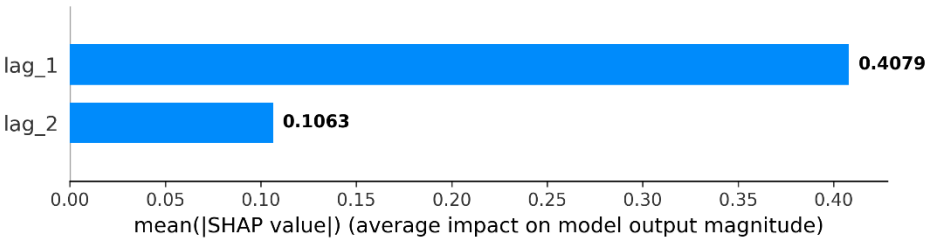
The market's *moderate optimism* regarding the annual GDP rate in 2026 is a mere 0.11 p.p. away from a model of paralysis. Consequently, the word *resilience* becomes nothing more than a euphemism, as to be resilient, an object or situation must, at the very least, return to its original state. What we have observed is a 0.20 p.p. retreat in the 2021-2025 period relative to 1996-2008. That is to say, the inertial level is worsening, drifting toward stagnation.

APPENDIX

The charts below expose why breaking the inertial ceiling is a nearly Herculean task under the current structure.

Appendix Chart 1 (SHAP Values) reveals that GDP (t-1) has a weight nearly four times greater than the previous quarter (t-2). This confirms that the Brazilian economy is "addicted" to its previous state, with no new productivity or investment vectors strong enough to deviate from the trajectory dictated by the past quarter.

Appendix Chart 1 – Feature Importance (SHAP)



Appendix Chart 2 is the visual proof of the trap. The critical mass of data (88.2% of the sample) is compressed into the central low-growth cluster (0.67%). Expansion zones appear as isolated, distant islands, while the system's gravity pulls any attempt at acceleration back into the Stagnation Zone. It is the geometric representation of a country that has forgotten how to grow.

Appendix Chart 2 – Structural Inertia Map: Decision Boundaries

